

Human Factors Risk Assessment

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Introduction

Commitment to both extensive and short-term space missions requires great rigor and dedication, particularly in the form of attention and care to make sure the station's accommodations and tools are well maintained. The overwhelming amount of equipment and machinery necessary for the proper function of the station demands much attention from the crew in orbit and on the ground. If gone unchecked or misused, this machinery and equipment could be damaged in forms such as water and food, as well as fuel tanks and circuitry. To provide the utmost protection to the onboard crew, this assessment will analyze three prominent hazards with their magnitude of danger, as well as their probability of going defective, and provide potential strategies or solutions to increase the crew's safety.

Risk Assessment

The first of the hazards that will be analyzed is the environment and atmospheric factors on the crew. One of the most prominent risks to the crew's health would be a defect in the mission necessary for carbon dioxide removal. A brief article by NASA states that the removal of carbon dioxide from the atmosphere is a challenge. This is because, unlike the natural processes on earth, which regulate the carbon dioxide levels, the crew must rely on carbon dioxide removal assembly, lithium hydroxide, and amine systems to regulate carbon dioxide exposure (Lewis, 2023). The USDA Food Safety and Inspection Service lists many health violations and dangers of excess CO₂ levels, the most critical to human health being high blood pressure, arrhythmias, and death by suffocation (USDA, n.d.). Therefore, the limitation of Carbon dioxide is crucial to the lives of the crew and for the prevention of jeopardizing space missions. For this reason, the environment and atmospheric risks for a space mission will receive a score of 5 for impact. However, a report by NASA on the testing of the carbon dioxide removal assembly concludes that the machinery operated flawlessly for each rendition and draft of testing (Knox, n.d.). For this reason, the reliability of the environmental systems of the ISS will receive a score of 2 on risk probability. It will not be given a lower score as risks may always emerge from human error when caring for the equipment, which is limited to the training effectiveness of the crew.

Secondly, the assessment will critique the safety and potential impact of the technology and its usability by the crew to determine where it is lacking, and identify the potential sources of the risk. A critical component of having a cooperative and productive crew on a space station is heavily reliant on clear and thorough communication with teams on Earth. Because of the inefficiency in work rate, which can be caused by miscommunication or a lack thereof, communication systems are almost always implemented into any space vessels to allow for coherent planning and alignment of ideas. Generally, teams on Earth are able to provide instructions and information to aid crew members on board and allow the team to function smoothly. An article by NASA states, a significant risk that can occur in the instance of a communications systems blackout is isolation of crew members from each other; too long of isolation can cause inadequate behavior for the crew members and lead to conflict on board (Lewis, 2023). This shows how, without proper communication with teams on Earth, crew in orbit can have inner conflict and lead to a jeopardization of the mission. For this reason, the communication system blackout will be given an impact of 3 on a scale of 0-5. Determining the probability of a communication system's failure requires an analysis of system failures, the reliability of backup systems, and the likelihood of environmental interference with communication signals. According to an article published by professors of computer engineering and physics, the probability of communication shortages is proportional to the mission duration and distance from Earth (Arnon and Gill, 2014). The source of this issue is signal degradation, with a farther distance, the signal will not be able to span for a longer period of time and for increasing distances. However, recent advancements in reliable communications systems such as laser communications limit the risk probability. To conclude, the probability of risk will be given a 3, because it is being limited by recent advancement, but still implies a large risk on mission success if it is to occur.

Thirdly, it is without question that the health and function of crew members should be one of, if not the highest priority of a space mission. Injuries of crew members are a dire deficit not only to the crew who need assistance, but to the success of the mission. The EVA missions could be referenced and analyzed in order to identify where the main causes of the injuries come from. An evidence report from NASA's human research road map, the most prominent types of injuries endured by astronauts were to the lower body, including ankles and lower back, as well as biceps, hands and forearms. The report cites that the injuries can be a result of repetitive demanding operations and tasks that weaken and strain the area overtime. Not only this,

studies within the report suggest that scenarios where astronauts must work with partial gravity require more physical effort by the muscles, compared to simply using the upper body in microgravity (Dunn, et al., 2022). The impact of these strains and injuries can over time degrade the body's work output, leading to reduced mission effectiveness. This could significantly hinder the crew's longevity of work on missions and delay waiting for the results and data the missions were conducted for. Therefore, the impact of human injury will receive a rating of 4 on a scale of 0-5 as it is not only an ethical impact, but also an impact on the mission's success and the performance of other crewmembers who need assistance. The report has statistics which state that out of the 219 in-flight injuries, 50 of them came from wearing the EVA suit. Partnered with the fact that it says that in the upcoming future, the hours of working on EVAs will increase to 24 hours of EVAs per week per astronaut (Dunn, et al., 2022). The probability of crewmate injury will certainly increase if action is not taken to mitigate the factors. Therefore, the risk probability will receive a score of 4 because it indicates an increase and prominent frequency.

Proposed Solutions

To begin, the proposed solutions will be reflective of the causes of and factors contributing to the risks assessed within the document. The first proposed solution will be in the form of a backup or alternative option for crew members in the case of a carbon dioxide removal assembly. Due to the low error and flawless output of the machines, it is recommended that the crew reserve a second machine to serve as a backup in case the other does not withstand improper use or misuse by the crew. This way, the crew's health and the mission's success will not be jeopardized by a severe spike in carbon dioxide levels within the station. Another significant change would be to complement a carbon scrubber. Carbon Scrubbers act as a remover of CO₂ and maintain a healthy environment, which can temporarily limit the harmful effects of CO₂ spikes. Moreover, to mitigate the risks of communication system blackouts or a loss in communication altogether, leveraging more communication channels will help have alternate connections in order to communicate. Another way is to utilize the low-orbit satellite constellations such as SpaceX's Starlink or Oneweb, which can cover a larger range of communications coverage. Additionally, an approach to minimizing the frequency of injuries to astronauts would be to develop tools and machinery for operations that are currently being done manually, either on the shuttle in orbit or with equipment on a celestial body. Furthermore, shifting more attention to improving the mobility of suits would enable astronauts to use less effort in movement, relieving strain on the upper and lower parts of the body.

Conclusion

By analyzing the risks using an impact versus probability matrix, a ranking can be made on which risks should be most prioritized and focused on improving. The atmospheric risks of the crew and communication systems shut down are both equal as a medium, while the injury of crew members ranks at a high. In conclusion, the safety of crew members and mitigating defects in equipment requires heavy commitment for care and maintenance, as well as a desire to develop new machinery that can handle manual tasks and improve already existing equipment to limit risks. By implementing backups, alternatives, and new technologies on board, space missions can be more efficient and ensure success. These proposed strategies not only ensure a safer health and well-being for the crew, but also cater to the longevity of the spacecraft. Effective preparations and action are necessary to maintain integrity in the future of space exploration, but also increase the likelihood of success, and the development of humanity's understanding of the universe and progress to achieving a better envisioned society.

Works Cited

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